

a Each validation target supports one claim, not the next

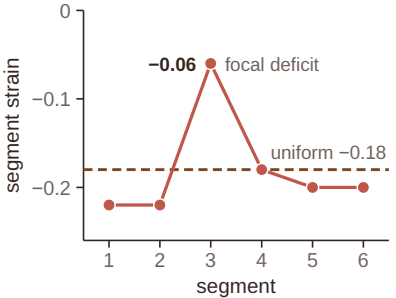
● Validation target	✓ Strongest claim it supports	✗ Does not establish on its own
Known motion phantom, STRAUS, MRXCAT2.0	representation capacity and technical error in the simulated regime	in-vivo accuracy
Paired DENSE or tagging same session, registered	material-sensitive agreement under matched definitions	focal location or extent, unless measured
Scan-rescan same subject, re-imaged	precision; accuracy needs a reference	bias
LGE or tissue characterisation scar or fibrosis label	biological association; motion error stays unmeasured	pointwise displacement error
Outcome events, prognosis	prognostic association; utility needs a decision study	decision impact

b Pre-specify, then escalate the reference stage by stage

Stage	What is varied or fixed	Resources (Table 4)
Pre-specify measurement contract	estimand, reference, tolerance and data splits fixed in advance	
Stage 1 Known motion	vary width, extent, amplitude, timing, resolution and noise	STRAUS MRXCAT2.0 CMAC 3 templates × 6 states programmable generator dynamic phantom
Stage 2 Paired material reference	cine with DENSE or tagging in one session; registered; reference blinded	Stanford CMAC 51/55 · not file-verified 15 volunteers, 12 landmarks
Stage 3 Tissue and decisions	LGE, responsiveness, decision impact	Clinical cohorts LGE or outcomes

Counts are resource units, not independent test participants.

c Same mean, focal deficit



segments 1–3, focal	-0.22 · -0.22 · -0.06
segments 4–6, focal	-0.18 · -0.20 · -0.20
global mean, both fields	-0.18
range, focal / uniform	0.16 / 0

Recovering the mean proves